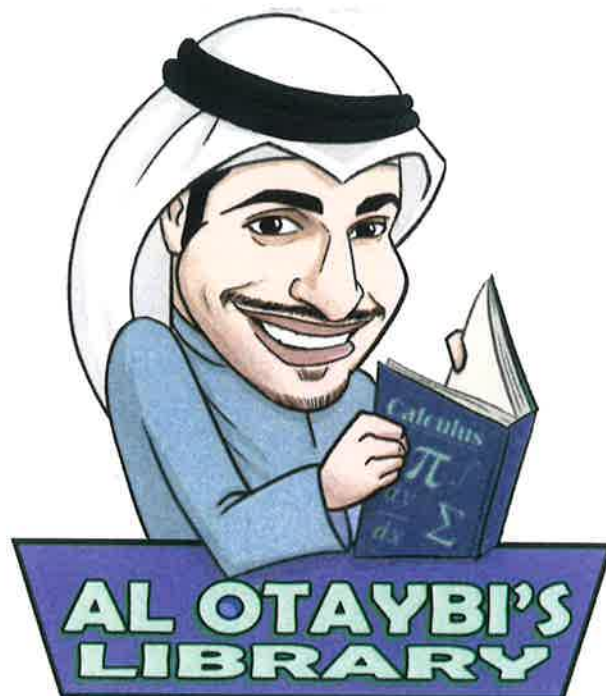


Kuwait University

2021/2022

Fall course



Calculus A

First

Review + Section 1.2 (part a)

Mathematical Models : A Catalog of Essential
Functions

(1)



Review



العامل المشترك



$$x^2 + 9x = x(x + 9)$$

$$3x^2 - 2x^3 = x^2(3 - 2x)$$

$$2x^2 - 4x = 2x(x - 2)$$

$$x - 9x^{10} = x(1 - 9x^9)$$



فرق بين مربعين



$$a^2 - b^2 = (a - b)(a + b)$$

$$x^2 - 9 = (x - 3)(x + 3)$$

$$4 - x^2 = (2 - x)(2 + x)$$

$$x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$$

$$4x^2 - 9 = (2x - 3)(2x + 3)$$

$$x^2 + 9$$

لا يمكن تحليلها





فرق أو مجموع مكعبين



$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$$

$$x^3 + 1 = (x + 1)(x^2 - x + 1)$$

$$27x^3 - 8 = (3x - 2)(9x^2 + 6x + 4)$$



المرافق



طريقة الضرب بحد والمرافق له.

مربع الأول (ناقص) مربع الثاني



$$(\sqrt{x} + 3)(\sqrt{x} - 3) = x - 9$$

$$\begin{aligned} (\sqrt{x+2} - 4)(\sqrt{x+2} + 4) &= x + 2 - 16 \\ &= x - 14 \end{aligned}$$

$$\begin{aligned} (5 - \sqrt{x-2})(5 + \sqrt{x-2}) &= 25 - (x - 2) \\ &= 25 - x + 2 \\ &= 27 - x \end{aligned}$$



تحليل حدودية من الدرجة الثانية



$$ax^2 \pm bx \pm c$$



$$x^2 - x - 2$$



$$(x+1)(x-2)$$

$$x^2 - 5x - 6$$

$$(x+1)(x-6)$$

$$x^2 + 2x - 3$$

$$(x-1)(x+3)$$

$$x^2 - 6x - 7$$

$$(x+1)(x-7)$$

$$x^2 + 4x - 12$$

$$(x-2)(x+6)$$

$$x^2 + 7x + 6$$

$$(x+1)(x+6)$$

$$x^2 + 6x + 5$$

$$(x+1)(x+5)$$

$$x^2 - 4x + 3$$

$$(x-1)(x-3)$$

$$x^2 - 3x + 2$$

$$(x-1)(x-2)$$



$$2x^2 - 3x + 1$$



$$(2x-1)(x-1)$$

$$3x^2 - 7x + 2$$

$$(3x-1)(x-2)$$

$$3x^2 - 11x + 6$$

$$(3x-2)(x-3)$$



توحيد المقامات



لا يمكن جمع أو طرح أي كسرين ما لم يكن المقام موحد



يجب توحيد المقامات في البداية

$$\frac{a}{b} \pm \frac{c}{d}$$

$$\frac{1}{x-1} + \frac{x}{x+2}$$

$$\frac{x+2}{(x-1)(x+2)} + \frac{x(x-1)}{(x+2)(x-1)} = \frac{x+2 + x(x-1)}{(x-1)(x+2)}$$

$$= \frac{\cancel{x} + 2 + \cancel{x^2} - \cancel{x}}{(x-1)(x+2)} = \frac{x^2 + 2}{(x-1)(x+2)}$$

$$\frac{1}{x-1} - \frac{x}{x+2}$$

$$\frac{x+2}{(x-1)(x+2)} - \frac{x(x-1)}{(x+2)(x-1)} = \frac{x+2 - x(x-1)}{(x-1)(x+2)}$$

$$= \frac{x+2 - x^2 + x}{(x-1)(x+2)} = \frac{-x^2 + 2x + 2}{(x-1)(x+2)}$$

$$\frac{1}{x-1} + \frac{1}{x^2-1}$$

$$\frac{1}{x-1} + \frac{1}{(x-1)(x+1)}$$

$$\frac{x+1}{(x-1)(x+1)} + \frac{1}{(x-1)(x+1)}$$

$$\frac{x+2}{(x-1)(x+1)}$$

$$\frac{x^2-1}{(x-1)(x^2-1)} + \frac{(x-1)}{(x^2-1)(x-1)}$$

$$\frac{x^2-1 + x-1}{(x-1)(x^2-1)}$$

$$\frac{x^2 + x - 2}{(x-1)(x^2-1)}$$

$$\frac{\cancel{(x-1)}(x+2)}{\cancel{(x-1)}(x-1)(x+1)}$$

$$\frac{x+2}{(x-1)(x+1)}$$



أحلى



Solve for x

$$x^2 + 4x = 0$$

$$x(x+4) = 0$$

$$x = 0, \quad x+4 = 0$$

$$x = -4$$

Solve for x

$$x^2 - x - 2 = 0$$

$$(x+1)(x-2) = 0$$

$$x+1 = 0, \quad x-2 = 0$$

$$x = -1$$

$$x = 2$$

Solve for x

$$x^2 - 4 = 0$$

$$(x-2)(x+2) = 0$$

$$x-2 = 0, \quad x+2 = 0$$

$$x = 2$$

$$x = -2$$

Sol #2

$$x^2 - 4 = 0$$

$$x^2 = 4$$

$$x = \pm 2$$



القانون العام لحل أي معادلة تربيعية



المميز



If $ax^2 + bx + c = 0$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



Solve for x

$$2x^2 - 5x + 1 = 0$$

$$a = 2$$

$$b = -5$$

$$c = 1$$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(1)}}{2(2)}$$

$$= \frac{5 \pm \sqrt{25 - 8}}{4}$$

$$= \frac{5 \pm \sqrt{17}}{4}$$



$$\therefore x = \frac{5 + \sqrt{17}}{4}, \quad x = \frac{5 - \sqrt{17}}{4}$$

Solve for x

$$2x^2 = 3 - 5x$$

$$2x^2 + 5x - 3 = 0$$

لازم تكون
مفرقة

$$a = 2 \quad b = 5 \quad c = -3$$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-5 \pm \sqrt{25 - 4(2)(-3)}}{2(2)}$$

$$= \frac{-5 \pm \sqrt{25 + 24}}{4}$$

$$= \frac{-5 \pm \sqrt{49}}{4}$$

$$= \frac{-5 \pm 7}{4}$$

$$x = \frac{-5 + 7}{4} \\ = \frac{2}{4} = \frac{1}{2}$$

$$x = \frac{-5 - 7}{4} \\ x = \frac{-12}{4} = -3$$

Solve for x

$$x = 3 + \frac{2}{x}$$

* x

لتبسيط المعادلة

$$x^2 = 3x + 2$$

$$x^2 - 3x - 2 = 0$$

$$a = 1 \quad b = -3 \quad c = -2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{3 \pm \sqrt{9 - 4(1)(-2)}}{2(1)}$$

$$\Rightarrow x = \frac{3 \pm \sqrt{17}}{2}$$



Types of functions



1) Polynomials

كل الاكسات أسهم صحيح و موجب



Examples

$y = 3x + 2$	→	polynomial of degree 1 (Linear function)
$f(x) = 4x^2 - 5x + 3$	→	polynomial of degree 2 (Quadratic function)
$g(x) = x^3 - 2x^2 + x - \sqrt{2}$	→	polynomial of degree 3 (Cubic function)
$y = 4x^2 - x^5$	→	polynomial of degree 5



2) Rational

Poly. حدودية
Poly. حدودية



Examples

$$y = \frac{x^2 - 1}{x + 2}$$

$$y = \frac{x + 3}{x^3 - 2}$$



3) Power

$$y = x^a$$



Examples

$$y = x^3$$

$$f(x) = x^2$$



استثناءات



Root

$$y = \sqrt[n]{x} = x^{\frac{1}{n}}$$

Reciprocal

$$y = \frac{1}{x}$$



Examples

$$y = \sqrt{x}$$

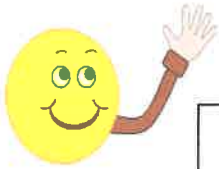
$$f(x) = \sqrt[3]{x}$$



4) Algebraic

كل جذر الحدوديات التي تحتوي على

جمع أو طرح أو ضرب أو قسمة



Examples

$$y = \frac{\sqrt{x} + 1}{x^2 - 3}$$

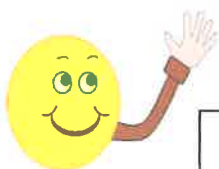
$$f(x) = \frac{\sqrt{x^2 - 4}}{x + 3}$$

$$y = \sqrt{x^2 + 3}$$



5) Trigonometric functions

الدوال المثلثية



Examples

$$y = \sin x$$

$$y = \cos x$$

$$y = \tan x$$

$$y = \csc x$$

$$y = \sec x$$

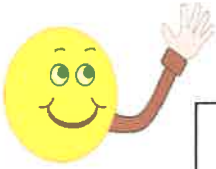
$$y = \cot x$$

ومقلوباتها



6) Exponential functions

$$y = a^x$$



Examples

$$y = 3^x \quad f(x) = 9^x \quad y = \left(\frac{1}{2}\right)^x$$



استثناء



Natural Exponential function

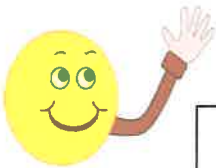
$$y = e^x = \exp(x), \quad e \approx 2.718$$



7) Logarithmic functions

$$y = \log_a x$$

رقم أكبر من الصفر ولا يساوي الواحد



Examples

$$y = \log_3 x \quad f(x) = \log_7 x \quad y = \log_1 x$$



لا يمكن للأساس
أن يكون الرقم 1



استثناء



Natural Logarithmic function

$$y = \ln x$$



8) inverse of trigonometric functions



Examples

$$y = \sin^{-1} x$$

$$y = \cos^{-1} x$$

$$y = \tan^{-1} x$$



ليست مقلوبات لهم

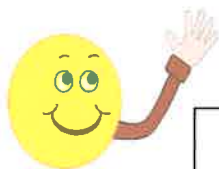
$$y = \csc^{-1} x$$

$$y = \sec^{-1} x$$

$$y = \cot^{-1} x$$



9) Hyperbolic functions



Examples

$$y = \sinh x$$

$$y = \cosh x$$

$$y = \tanh x$$



ومقلوباتها

$$y = \operatorname{csch} x$$

$$y = \operatorname{sech} x$$

$$y = \operatorname{coth} x$$



سوف يتم التطرق لهم بشكل
مفصل في هذا المقرر





Text book question



Classify each function as power function , root function , polynomial (state the degree) , rational function , algebraic function , trigonometric function , exponential function or logarithmic function.

a) $y = \pi^x$



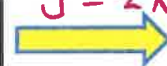
exponential function

b) $y = x^\pi$



power function

c) $y = x^2(2 - x^3)$



$y = 2x^2 - x^5$

polynomial of degree 5

d) $y = \tan t - \cos t$



trigonometric function

e) $y = \frac{s}{1+s}$



rational function

f) $y = \frac{\sqrt{x^3 - 1}}{1 + \sqrt[3]{x}}$



algebraic function



Text book question



Classify each function as power function , root function , polynomial (state the degree) , rational function , algebraic function , trigonometric function , exponential function or logarithmic function.

a) $f(x) = \log_2 x$



logarithmic function

b) $g(x) = \sqrt[4]{x}$



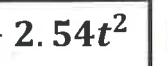
root function

c) $h(x) = \frac{2x^3}{1 - x^2}$



rational function

d) $u(t) = 1 - 1.1t + 2.54t^2$



Polynomial of degree 2
Quadratic function

c) $v(t) = 5^t$



exponential function

f) $w(\theta) = \sin \theta \cos^2 \theta$



trigonometric function